

Maciej Karczmarz

Go / Cloud / DevOps | Computer Science Student (Rzeszów University of Technology)

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Summary

Computer Science Master's student (Rzeszów University of Technology) with experience in Go, Python and cloud infrastructure (AWS, Kubernetes, GitOps). I build backends and automations, from a quoting system for a business client to a platform that turns LiDAR data into OpenUSD scenes. Looking for an internship, working student or part-time role in DevOps / cloud / Go backend that fits around my studies (remote or hybrid).

Skills

Languages & data:
Go, Python, Shell, TypeScript | PostgreSQL (TimescaleDB, PostGIS, pgvector), Redis, NATS, gRPC

Cloud / DevOps / 3D:
AWS, Kubernetes, Helm, GitOps, Docker, Datadog, n8n; Linux/macOS. | Blender, Unreal Engine 5, Houdini, OpenUSD

Projects

Applications/Infrastructure

Nequ3D (master's thesis): Platform for processing LiDAR and photogrammetry data into optimized OpenUSD scenes (texture compression with NVIDIA RTX NTC). Asynchronous Go backend, containerized Python workers, Wails frontend with local LLM reporting, Kubernetes infrastructure managed via GitOps.
Technologies: Go, Python, Docker, Kubernetes, GitOps, OpenUSD, NVIDIA RTX-NTC

SmartOffer (competition, client AB Bechcicki): AI agent automating B2B quoting in construction: from conversation, voice, or photos and attachments it generates print-ready quotes from a catalog of ~98k products (PostgreSQL + pgvector). Agent orchestration in n8n, Next.js frontend, deployed with Docker on AWS EC2. Team project.
Technologies: TypeScript, Next.js, n8n, PostgreSQL, pgvector, Docker, AWS

Gix: Distributed currency exchange rate monitor for Polish exchange offices: scraping and aggregation, real-time streaming to a native desktop client (Gio). Go (Gin) backend, NATS JetStream, dRPC, Redis, TimescaleDB; deployed on Kubernetes (DigitalOcean, Helm), observability with Datadog. Built-in scraper cost estimation (FinOps, work in progress).
Technologies: Go, NATS, Redis, TimescaleDB, Kubernetes, Helm, Datadog

Oentike: Mushroom-conditions helper for Polish forests: a 0–100 score computed from weather data (Open-Meteo) and forest district data (BDL), Go gRPC API with hourly background ingest, PostGIS, Tauri desktop app.
Technologies: Go, gRPC, PostGIS, Tauri

Other projects: Honeypot (Shell), Sand-Simulation (C++, SFML), Unreal Engine 5 environment (Gaea 2.0, Houdini)

Experience

Simplicity Games Rzeszow, Poland
Technical Artist November 2024 – January 2025
3D model optimization (Blender: low-poly, LODs, collision meshes) and PBR textures for performance and Unity engine requirements.
https://simplicitygames.pl/

Education

Rzeszow University of Technology ~expected
Computer Science March 2026 – July 2027
Master's studies – Faculty of Electrical and Computer Engineering

Rzeszow University of Technology October 2022 – February 2026
Computer Science
Bachelor's (Engineering) studies, Faculty of Electrical and Computer Engineering

- Projects: graphical simulation in C++, data analysis (Python, MATLAB, R), team web app in React, AI project (MLP). Hackathons (incl. NASA Space Apps Challenge, HackCarpathia), student research group SKNI KOD since 2022.

Languages

Polish	English	German
Native	C1	A2